Sample Document for AI Question Answering

A short practice PDF for upload, extraction, chunking, embedding, and retrieval tests.

1. Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science focused on building systems that perform tasks normally associated with human intelligence. Examples include language understanding, image recognition, planning, and decision support.

2. Machine Learning

Machine learning (ML) is a subset of AI. It enables a system to learn patterns from examples rather than relying only on hand-written rules. Common learning types include supervised learning, unsupervised learning, and reinforcement learning.

3. Embeddings and Vector Search

An embedding represents text as a list of numbers called a vector. Texts with similar meanings can have vectors that are close in an embedding space. Vector search compares vectors to retrieve document passages relevant to a user's question.

4. Retrieval-Augmented Question Answering

A document question answering system extracts text from uploaded documents, divides long text into chunks, creates embeddings, retrieves relevant chunks, and uses those chunks as context for answer generation. The system should show supporting source text and should say when the answer is not found in the documents.

5. Example Facts for Testing

The fictional test project is named PaperPilot. It uses a chunk size of 800 characters and an overlap of 120 characters for its initial experiment. The embedding model used in this example is all-MiniLM-L6-v2. These settings are illustrative, not universal best practices.

Suggested test questions

• What is machine learning? • What does an embedding represent? • Why does the system split text into chunks? • What is the fictional project called? • Which chunk size is used in the example? • What should the system do if the answer is not in the document?